

R&J WOOD TRADING

VITREX · Tropical Hardwood & Botanical Sourcing

MATERIALS ENGINEERING BRIEF

Merbau vs. Balau

Load-bearing performance of two heavy tropical hardwoods for commercial and marine decking

PREPARED FOR

Architects · Structural Engineers · Commercial Decking Contractors

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1. Introduction: The Demands of Commercial and Marine Decking

Commercial decking is a structural problem before it is an aesthetic one. A boardwalk, a marina pontoon, a rooftop terrace over occupied space, or a high-traffic public promenade must carry distributed live loads and concentrated point loads, resist fatigue over decades of foot traffic, and survive a service environment that combines moisture cycling, ultraviolet exposure, fungal decay pressure and — in tropical and subtropical settings — relentless termite and marine-borer attack.

Two Southeast Asian heavy hardwoods dominate the specifier's shortlist for these conditions: **Merbau** (*Intsia palembanica*, and the closely related *I. bijuga*) and **Balau** (a trade group of dense *Shorea* species). Both are Class-durable, both are dense enough to deter most borers, and both have decades of field history. They are not, however, interchangeable. This paper compares them on the properties that govern a structural decking decision — Janka hardness, modulus of rupture (MOR), modulus of elasticity (MOE), density and dimensional behaviour — and translates those numbers into design guidance.

A note on nomenclature before we quote numbers. “Balau” is a trade name covering many *Shorea* species (Yellow Balau, Red Balau, Bangkirai, Selangan Batu), whose properties vary across a band rather than sitting at a single point. “Merbau” is botanically tighter (*Intsia* spp.) and therefore more consistent lot-to-lot. Any single published figure for Balau is a representative value within a range — engineers should design to the conservative end and verify the supplied species.

2. Species Profiles

2.1 Merbau — *Intsia palembanica*

Family Fabaceae (Leguminosae). Heartwood is yellow- to orange-brown when freshly cut, maturing to a deep red-brown; the grain is typically interlocked with a coarse but even texture, and the pores frequently carry a yellow deposit that finishes with a characteristic gold flecking. Merbau is prized for **exceptional dimensional stability** — often compared to teak — which is precisely why it performs so reliably as wide-board decking that must stay flat across wet/dry cycles. It is highly resistant to termites. Two working cautions: it dulls and gums cutting edges (carbide tooling is recommended), and its tannins will leach and stain — and will react with ferrous metals to produce black staining — so stainless fixings and run-off management are non-negotiable on visible work.

2.2 Balau — *Shorea* spp.

Family Dipterocarpaceae. Within the dense *Shorea* group, the hardest, heaviest timbers — Balau, Selangan Batu, Yakal — sit at the top of the density ladder. Heartwood ranges from yellow-brown (Yellow Balau) to reddish- or purplish-brown (Red Balau), with straight-to-interlocked grain and a coarse texture. Balau is rated highly durable with excellent decay and insect resistance, and it is the workhorse exterior hardwood across much of Europe and Australia. It is hard to machine, has a blunting effect on cutters

(and some species contain trace silica), and — like all unfinished tropical hardwood — will silver to grey under UV if left unoiled, with some surface checking in high-sun exposure.

Sustainability flag for specifiers. Neither genus is CITES-listed, but many *Shorea* species appear on the IUCN Red List, and species substitution within the “Balau” trade name is common. Merbau has also seen significant wild-population pressure in parts of its range. For both species, certified, traceable supply is now a procurement requirement as much as an ethical one — see our companion EUDR briefing.

3. Mechanical Properties: Side by Side

The table below consolidates representative published values at approximately 12% moisture content. Balau figures are given as a band reflecting the species spread within the trade group; Merbau figures are tighter. All values should be treated as indicative for preliminary design and confirmed against project-specific test data or a recognised timber-strength standard before final structural sizing.

Property	Merbau (<i>Intsia palembanica</i>)	Balau (<i>Shorea spp.</i>)	Engineering relevance
Air-dry density	≈ 800–840 kg/m ³	≈ 830–1,070 kg/m ³	Mass, fixing & substructure loads
Specific gravity	≈ 0.80–0.83	≈ 0.75–0.85	Proxy for strength & wear
Janka hardness	≈ 1,710 lbf (≈ 9.0 kN)	≈ 1,600–1,700+ lbf (≈ 7.1 kN+)	Dent / wear resistance
Modulus of rupture (MOR)	≈ 147 MPa	≈ 120–172 MPa	Bending (failure) strength
Modulus of elasticity (MOE)	≈ 18 GPa	≈ 16–18 GPa	Stiffness — deflection control
Radial shrinkage	≈ 1.5%	≈ 4.2%	Movement across the board
Tangential shrinkage	≈ 2.5%	≈ 6.2%	Cupping / gapping tendency
Natural durability	Durable; termite-resistant	Highly durable; decay/insect-resistant	Service life untreated

Sources: *The Wood Database* (Merbau; Balau); *Britton Timbers species data* (Merbau); *Advantage Lumber technical data* (Yellow & Red Balau). Values at ~12% MC. Balau ranges reflect species variation within the trade group.

3.1 Reading the numbers

Strength and stiffness are close. On bending strength (MOR) and stiffness (MOE) the two timbers are broadly comparable. The densest Red Balau can exceed Merbau on raw MOR, but Merbau's figures are more *predictable* because the species is botanically tighter. For a designer, predictability is itself a structural virtue: it narrows the gap between the characteristic value used in design and the mean value of the timber actually delivered.

Hardness is effectively a tie. Both sit around 1,600–1,710 lbf on the Janka scale — comfortably past the threshold where additional hardness yields diminishing real-world wear benefit for a deck surface. Both will resist denting from furniture, trolleys and heavy footfall in commercial use.

Dimensional stability is where they part company. This is the most decision-relevant line in the table. Merbau's radial/tangential shrinkage (≈ 1.5% / 2.5%) is markedly lower than Balau's (≈ 4.2% / 6.2%). In service, that means Merbau moves less across its width with humidity swings — less cupping, less

gapping, fewer fastener problems, and a flatter long-term surface. For wide boards, for fully-exposed decks with large wet/dry swings, and for marine splash zones, this is a significant advantage.

4. Durability and Weathering Over 20+ Years

Both species are classed as durable to highly durable, and both will deliver multi-decade service in exterior use when detailed correctly. The differences emerge in how they age and what failure modes dominate.

Decay and biological attack

Balau carries an explicit “highly durable” rating with excellent resistance to both decay fungi and insects, and a long European track record in fully-weathered applications. Merbau is rated durable and notably termite-resistant. In ground-contact or permanently-wet detailing, both should be specified with the durability of the heartwood only — sapwood of either species is not durable and should be excluded.

Movement, checking and surface life

Here Merbau’s low shrinkage pays back over the service life: less seasonal cupping and a more stable surface mean fewer trip hazards and a slower onset of surface checking. Balau, with higher movement, benefits more from disciplined detailing — generous board gaps, good airflow beneath, effective drainage and a maintained oil finish — to manage checking and silvering. Neither timber should be installed without back-ventilation.

The fixings caveat — especially Merbau

Merbau’s tannins leach freely in early weathering and will stain adjacent light-coloured surfaces (render, stone, pale paving) brown; in contact with iron or ordinary steel they cause black staining. Stainless-steel fixings, tannin-managed run-off and an initial weathering/wash regime are essential where appearance matters. Balau’s leaching is less aggressive but stainless fixings remain best practice for both.

Marine environments. For pontoons, jetties and splash-zone work, dimensional stability and movement control become the governing concern alongside durability. Merbau’s low shrinkage makes it the more forgiving choice for wide marine decking and boat-side fit-out; Balau performs well where boards are narrower, well-gapped and ventilated, and where the budget favours its broader availability.

5. Recommendation by Project Type

Both species are correct answers for heavy-duty exterior decking. The right choice is driven by board width, exposure, appearance tolerance and budget rather than by any large gap in raw strength.

Application	Recommended	Rationale
Wide-board commercial / rooftop decking	Merbau	Low movement keeps wide boards flat; stable surface
Marine pontoons & splash-zone fit-out	Merbau	Dimensional stability governs; termite + moisture resistance
High-traffic public promenade	Either	Both exceed wear thresholds; pick on budget & supply
Narrow-board, ventilated exterior decking	Balau	Excellent durability; cost-effective when detailed well
Large-area projects on tight budgets	Balau	Broader availability and typically lower cost
Heaviest point-load substructure members	Red Balau	Densest grades give highest MOR where mass is acceptable

In short: specify **Merbau** when dimensional stability, wide boards or a stain-managed visible surface dominate the brief; specify **Balau** when you want a highly durable, well-priced, widely available hardwood and you can detail for its greater movement. In every case, design to characteristic values for the confirmed species, exclude sapwood, fix in stainless, and ventilate beneath.

Engineering disclaimer. Published mechanical values are indicative and vary with growth conditions, moisture content, species within a trade group, and specimen. They are provided for preliminary comparison only and must not be used as characteristic design values. Confirm structural properties against project-specific testing or an applicable timber-engineering standard, and engage a qualified structural engineer for load-bearing design.

Key sources. The Wood Database (Merbau, Balau, Red/Yellow Balau); Britton Timbers (Merbau species data sheet); Advantage Lumber technical specifications. Values at approximately 12% moisture content.